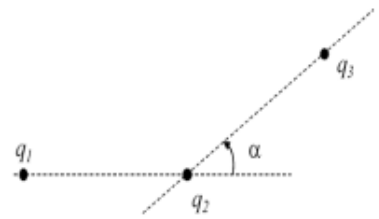


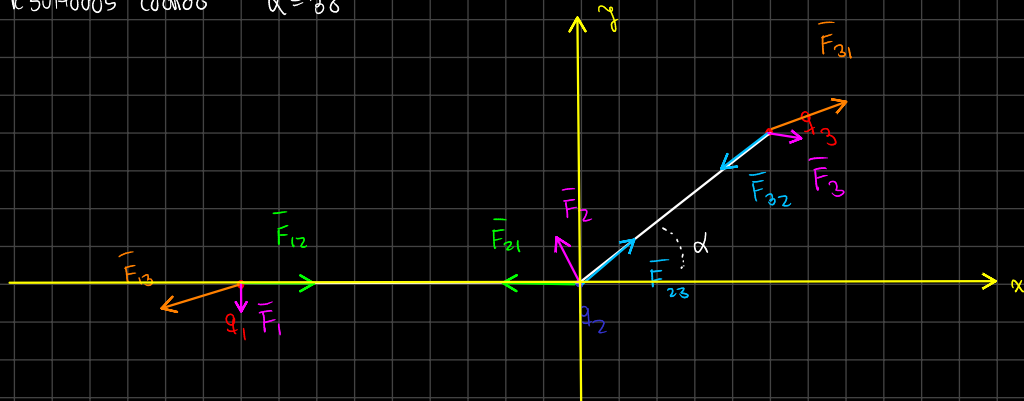
2. ¿Qué fuerza se debe ejercer sobre cada una de las tres cargas puntuales de la figura ($q_1 = 1 \mu\text{C}$, $q_2 = -2 \mu\text{C}$ y $q_3 = 0,5 \mu\text{C}$) para que el sistema esté en equilibrio?

La distancia entre q_1 y q_2 es igual a la distancia entre q_3 y q_2 y vale 30 cm. Analizar los resultados cuando el ángulo $\alpha = 30^\circ$.



Datos: $q_1 = 1 \mu\text{C}$ $q_2 = -2 \mu\text{C}$ $q_3 = 0,5 \mu\text{C}$ $d = 30 \text{ cm}$ $\alpha = 30^\circ$

Analizar los resultados cuando $\alpha = 30^\circ$



$$\vec{F}_1 + \vec{F}_2 + \vec{F}_3 = 0$$

$$\vec{F}_{12} + \vec{F}_{13} + \vec{F}_{21} + \vec{F}_{23} + \vec{F}_{31} + \vec{F}_{32} = 0$$

Solo debo calcular $\vec{F}_{12}$, $\vec{F}_{13}$ y $\vec{F}_{23}$

$$\vec{F}_{12} = k \frac{q_1 q_2 (\vec{r}_1 - \vec{r}_2)}{|\vec{r}_1 - \vec{r}_2|^3}$$

$$\vec{r}_1 = -d\hat{i} \quad \vec{r}_1 - \vec{r}_2 = -d\hat{i}$$

$$\vec{r}_2 = \vec{0} \quad |\vec{r}_1 - \vec{r}_2| = d$$

$$\vec{F}_{12} = k \frac{q_1 q_2 \cdot -d\hat{i}}{d^3}$$

$$\vec{F}_{12} = -k \frac{q_1 q_2}{d^2} \hat{i}$$

$$\vec{F}_{12} = -9 \cdot 10^9 \frac{\text{Nm}^2}{\text{C}^2} \times \frac{(1 \mu\text{C})(-2 \mu\text{C})}{(0,3 \text{ m})^2} \hat{i}$$

$$\vec{F}_{12} = -9 \cdot 10^9 \frac{\text{Nm}^2}{\text{C}^2} \times \frac{(1)(-2) \mu\text{C}^2}{(0,3)^3 \text{ m}^2} \hat{i}$$

$$\vec{F}_{12} = -9 \cdot 10^9 \text{ N} \times \frac{(-2)(10^{-6})^2}{(0,3)^2} \hat{i}$$

$$\vec{F}_{12} = 0,2 \text{ N} \hat{i}$$

$$\vec{F}_{23} = k q_2 q_3 \frac{(\vec{r}_2 - \vec{r}_3)}{|\vec{r}_2 - \vec{r}_3|^3}$$

$$\vec{F}_{23} = k q_2 q_3 \frac{(-d \cos(\alpha) \hat{i} - d \sin(\alpha) \hat{j})}{d^3}$$

$$\vec{F}_{23} = k q_2 q_3 \frac{(-\cos(\alpha) \hat{i} - \sin(\alpha) \hat{j})}{d^2}$$

$$\vec{F}_{23} = \frac{9 \cdot 10^9 \text{ Nm}^2}{\text{C}^2} \times (-2 \mu\text{C})(0.5 \mu\text{C}) \frac{(-\cos(30^\circ) \hat{i} - \sin(30^\circ) \hat{j})}{(0.3 \text{ m})^2}$$

$$\vec{F}_{23} = 0.0866 \text{ N} \hat{i} + 0.05 \text{ N} \hat{j}$$

Puedo calcular $\vec{F}_2 = \vec{F}_{21} + \vec{F}_{23} = -\vec{F}_{12} + \vec{F}_{23}$

$$= -0.2 \text{ N} \hat{i} + 0.0866 \text{ N} \hat{i} + 0.05 \text{ N} \hat{j}$$

$$\vec{F}_2 = -0.1134 \text{ N} \hat{i} + 0.05 \text{ N} \hat{j}$$

Ahora voy a calcular $\vec{F}_{13}$

$$\vec{F}_{13} = k q_1 q_3 \frac{(\vec{r}_1 - \vec{r}_3)}{|\vec{r}_1 - \vec{r}_3|^3}$$

$$\# \vec{r}_1 = -d \hat{i} \quad \# \vec{r}_3 = d \cos(\alpha) \hat{i} + d \sin(\alpha) \hat{j}$$

$$\# \vec{r}_1 - \vec{r}_3 = -d \hat{i} - d \cos(\alpha) \hat{i} - d \sin(\alpha) \hat{j} = (-d - d \cos(\alpha)) \hat{i} - d \sin(\alpha) \hat{j}$$

$$= -d(1 + \cos(\alpha)) \hat{i} - d \sin(\alpha) \hat{j}$$

$$\# |\vec{r}_1 - \vec{r}_3|^3 = \sqrt{d^2(1 + \cos(\alpha))^2 + d^2 \sin^2(\alpha)}$$

$$= \sqrt{d^2(1 + 2\cos(\alpha) + \cos^2(\alpha)) + d^2 \sin^2(\alpha)}$$

$$= \sqrt{d^2 + 2d^2 \cos(\alpha) + d^2 \cos^2(\alpha) + d^2 \sin^2(\alpha)}$$

$$= \sqrt{2d^2 + 2d^2 \cos(\alpha)}$$

$$= \sqrt{2d^2(1 + \cos(\alpha))}^3$$

$$q_1 = 1 \mu\text{C}$$

$$q_2 = -2 \mu\text{C}$$

$$q_3 = 0.5 \mu\text{C}$$

$$d = 30 \text{ cm}$$

$$\alpha = 30^\circ$$

$$\vec{F}_{13} = \frac{k q_1 q_3 \left[-d(1 + \cos(\alpha)) \hat{i} - d \sin(\alpha) \hat{j} \right]}{\left[2d^2(1 + \cos(\alpha)) \right]^{3/2}}$$

$$F_{13x} = \frac{k q_1 q_3 \left[-d(1 + \cos(\alpha)) \hat{i} \right]}{\left[2d^2(1 + \cos(\alpha)) \right]^{3/2}}$$

$$F_{13x} = \frac{9 \cdot 10^9 \frac{\text{Nm}^2}{\text{C}^2} \times (1 \mu\text{C})(0.5 \mu\text{C}) \left[-0.3 \text{ m}(1 + \cos(30^\circ)) \hat{i} \right]}{\left[2(0.3 \text{ m})^2(1 + \cos(30^\circ)) \right]^{3/2}} \} \rightarrow 0.19466375$$

$$F_{13x} = -0.01294 \text{ N } \hat{i} //$$

$$F_{13y} = \frac{k q_1 q_3 \left[-d \sin(\alpha) \hat{j} \right]}{\left[2d^2(1 + \cos(\alpha)) \right]^{3/2}}$$

$$F_{13y} = \frac{9 \cdot 10^9 \frac{\text{Nm}^2}{\text{C}^2} \times (1 \mu\text{C})(0.5 \mu\text{C}) \left[-0.3 \text{ m} \sin(30^\circ) \hat{j} \right]}{\left[2(0.3 \text{ m})^2(1 + \cos(30^\circ)) \right]^{3/2}} \} \rightarrow 0.19466375$$

$$F_{13y} = -0.0034675 \text{ N } \hat{j}$$

$$\vec{F}_{13} = -0.01294 \text{ N } \hat{i} - 0.0034675 \text{ N } \hat{j}$$

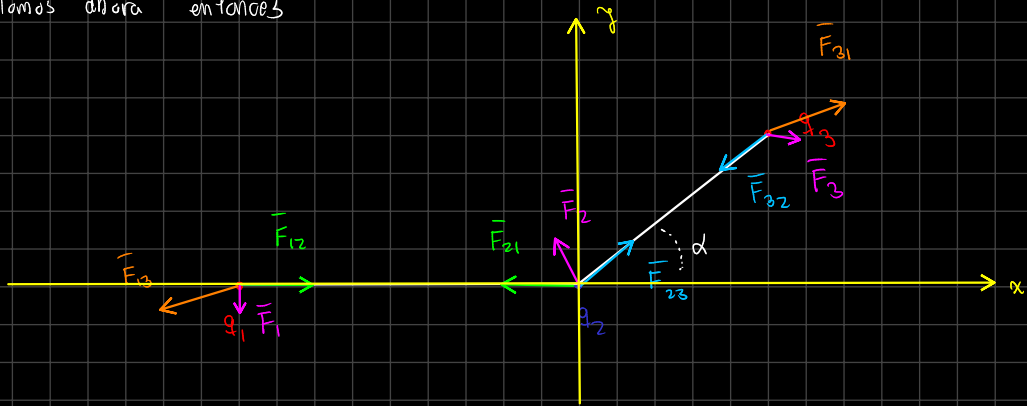
Podre Calcular $\vec{F}_1$

$$\vec{F}_1 = F_{12} + F_{13}$$

$$= 0.2 \text{ N } \hat{i} - 0.01294 \text{ N } \hat{i} - 0.0034675 \text{ N } \hat{j}$$

$$\vec{F}_1 = 0.18706 \text{ N } \hat{i} - 0.0034675 \text{ N } \hat{j}$$

Calculamos ahora entonces



$$\vec{F}_3 = \vec{F}_{31} + \vec{F}_{32} = -\vec{F}_{13} - \vec{F}_{23}$$

$$= 0.01294 \text{ N} \hat{i} + 0.0034675 \text{ N} \hat{j} - 0.0866 \text{ N} \hat{i} - 0.05 \text{ N} \hat{j}$$

$$= -0.07366 \text{ N} \hat{i} - 0.0465 \text{ N} \hat{j}$$

$$\boxed{\vec{F}_3 = -0.0736 \text{ N} \hat{i} - 0.0465 \text{ N} \hat{j}}$$